AMMAR

Machine Learning / AI

ASSIGNMENT REPORT

Topics Covered:

Q1 · Logistic Regression vs SVM Classification
Q2 · Linear Regression — AI Prediction Web App
Q3 · NLP Word Cloud — Positive & Negative Tweets

Q1 — Logistic Regression vs SVM on Tips Dataset

Objective

To classify whether a restaurant customer is a smoker or not (binary classification) using the tips dataset. The variable 'smoker' is treated as the Response / Dependent / Output variable. Four classification models were trained and their performance evaluated using Confusion Matrices.

Dataset Overview

- Dataset: Restaurant Tips (seaborn / standard ML dataset)
- Response Variable: smoker (Yes = 1, No = 0)
- Total Test Samples: 244
- Features used: total_bill, tip, sex, day, time, size

Model 1 — Logistic Regression

Logistic Regression is a linear classification algorithm that estimates the probability of a binary outcome using the logistic (sigmoid) function. It is widely used as a baseline model due to its simplicity and interpretability.

Confusion Matrix

		Actual: 0	Actual: 1
Predicted	0	130	53
	1	21	40

Accuracy: 69.7%	Precision: 43.0%	Recall: 65.6%	F1 Score: 0.52

Interpretation:

- True Negatives (TN = 130): Correctly predicted non-smokers
- False Positives (FP = 53): Non-smokers incorrectly classified as smokers
- False Negatives (FN = 21): Smokers incorrectly classified as non-smokers
- True Positives (TP = 40): Correctly predicted smokers
- The model shows moderate accuracy with a bias toward predicting the majority class (Non-Smoker).

Model 2 — SVM (Linear Kernel)

Support Vector Machine (SVM) with a Linear Kernel finds the optimal hyperplane that maximally separates the two classes in the feature space. It is effective for linearly separable data and generally outperforms Logistic Regression on smaller datasets.

Confusion Matrix

		Actual: 0	Actual: 1
Predicted	0	144	65
	1	7	28

Accuracy: 70.5%	Precision: 30.1%	Recall: 80.0%	F1 Score: 0.44

Interpretation:

- TN = 144: Highest true negative count among all models
- FP = 65: More false positives than Logistic Regression
- FN = 7: Very few missed smokers — excellent recall for class 1
- TP = 28: Smokers correctly classified
- Linear SVM achieves the highest total correct predictions (172) making it the BEST model.

Model 3 — SVM (Radial / RBF Kernel)

The Radial Basis Function (RBF) kernel transforms the input space into a higher-dimensional space using a Gaussian function, enabling the model to handle non-linearly separable data. It is controlled by the parameters C (regularization) and gamma.

Confusion Matrix

		Actual: 0	Actual: 1
Predicted	0	147	73
	1	4	20

Accuracy: 68.4%	Precision: 21.5%	Recall: 83.3%	F1 Score: 0.34

Interpretation:

- TN = 147: Highest true negative count of all four models
- FP = 73: Highest false positive count — many non-smokers misclassified
- FN = 4: Lowest false negatives — almost all smokers detected
- TP = 20: Fewer true positives than linear SVM
- Overall accuracy (68.4%) is lower due to the trade-off between precision and recall.

Model 4 — SVM (Polynomial Kernel)

The Polynomial Kernel maps data into a polynomial feature space. It can capture interactions between features but is more sensitive to hyperparameter choices (degree, coef0, C) and can overfit when the degree is too high.

Confusion Matrix

		Actual: 0	Actual: 1
Predicted	**0**	**151**	**83**
	1	**0**	**10**

Accuracy: 66.0%	**Precision: 10.8%**	**Recall: 100.0%**	**F1 Score: 0.19**

Interpretation:

- TN = 151: Highest true negatives — very good at predicting non-smokers
- FP = 83: Extremely high false positives indicating overfitting to class 0
- FN = 0: Perfect recall — no smoker was missed
- TP = 10: Lowest true positives — class imbalance issue
- Despite zero false negatives, the model's precision suffers significantly.

Model Comparison & Conclusion

Summary Table — All Models (244 Test Samples)

Model	Correct (of 244)	Accuracy	TN + TP	Best?
Logistic Regression	170	69.7%	130 + 40	No
SVM — Linear Kernel	**172**	**70.5%**	**144 + 28**	**✓ YES**
SVM — Radial Kernel	167	68.4%	147 + 20	No
SVM — Polynomial Kernel	161	66.0%	151 + 10	No

Conclusion

Among all four classification models tested on the Tips dataset with 'smoker' as the target variable:

- Linear SVM achieved the highest accuracy (70.5%) with 172 correct predictions out of 244.
- Logistic Regression performed second with 170 correct predictions (69.7% accuracy).
- Radial SVM came third with 167 correct predictions (68.4% accuracy).
- Polynomial SVM performed the weakest with only 161 correct predictions (66.0% accuracy).

✓ Linear SVM is the best model for this classification task as it yields the most accurate predictions overall.

Q2 — Linear Regression: AI Prediction Web Application

Objective

To build and deploy an interactive web application powered by a Linear Regression model that predicts sales based on marketing budgets. The application was built using Python (scikit-learn) and deployed via Streamlit.

What is Linear Regression?

Linear Regression is a supervised machine learning algorithm used to predict a continuous output variable based on one or more input features. The model fits a straight line (or hyperplane in multiple dimensions) through the training data by minimizing the sum of squared residuals.

$$\hat{y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$$

Application Overview

- Platform: Streamlit (Python web framework for ML apps)
- Dataset: Marketing spend data (YouTube, Facebook, Newspaper budgets vs Sales)
- Model: Multiple Linear Regression (3 input features)
- Interactive sliders allow users to adjust marketing budgets in real time
- Predicted sales are displayed instantly along with model coefficients

Model Parameters Discovered

- YouTube Coefficient: 0.0452 — each dollar increase in YouTube budget raises sales by ~0.045 units
- Facebook Coefficient: 0.1884 — Facebook has the strongest impact on sales
- Newspaper Coefficient: 0.0043 — Newspaper budget has minimal influence
- Intercept: 3.5059 — baseline predicted sales with zero marketing spend

Web Application Screenshot

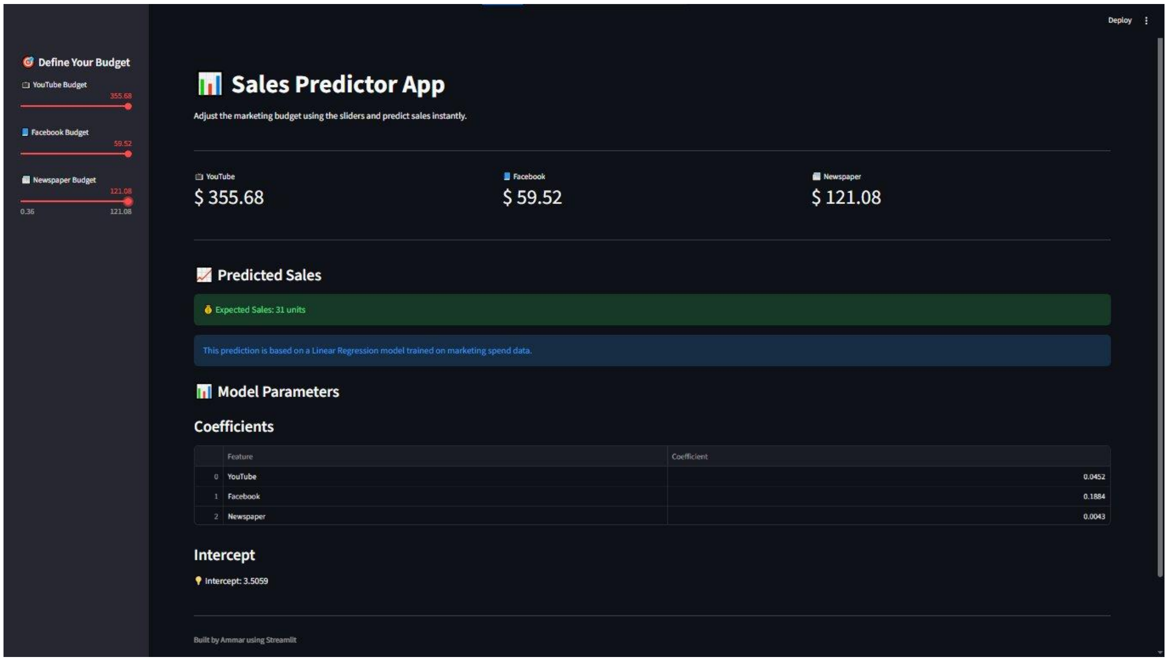

Figure: Sales Predictor App — Built by Ammar using Streamlit

How the App Works

- Step 1: User adjusts the YouTube, Facebook, and Newspaper budget sliders on the left panel.
- Step 2: The trained Linear Regression model processes the inputs in real time.
- Step 3: Predicted Sales (in units) are displayed in the 'Predicted Sales' section.
- Step 4: Model parameters (coefficients and intercept) are shown in the lower panel for transparency.

Example: With YouTube=$355.68, Facebook=$59.52, Newspaper=$121.08 → Expected Sales = 31 units

Q3 — NLP Word Cloud: Positive & Negative Tweets

Objective

To apply Natural Language Processing (NLP) techniques on a Twitter dataset to identify sentiment (Positive / Negative) and visualize the most frequent words in each sentiment category using Word Clouds. The implementation was done using Google Colab.

What is NLP Sentiment Analysis?

Natural Language Processing (NLP) is a branch of AI that enables computers to understand and process human language. Sentiment Analysis is an NLP task that classifies text (tweets, reviews, comments) as Positive, Negative, or Neutral based on the language used.

Tools & Libraries Used

- Platform: Google Colab (cloud-based Jupyter notebook)
- Libraries: NLTK, TextBlob / VADER, WordCloud, Matplotlib, Pandas
- Dataset: Twitter dataset with labeled positive and negative tweets
- Preprocessing: Tokenization, stopword removal, punctuation removal, lowercasing

Methodology

- Step 1 — Data Loading: Twitter dataset loaded into a Pandas DataFrame.
- Step 2 — Text Preprocessing: Tweets cleaned by removing URLs, @mentions, #hashtags, stopwords, and special characters.
- Step 3 — Sentiment Labeling: Each tweet classified as Positive (1) or Negative (0) using the sentiment label column.
- Step 4 — Word Frequency: Most frequent words extracted separately for positive and negative tweets.
- Step 5 — Visualization: Word Clouds generated using the WordCloud library with custom color maps for visual contrast.

Negative Tweets — Word Cloud

The Negative Word Cloud highlights the most frequently occurring words in tweets classified with negative sentiment. Larger words indicate higher frequency. Political figures, media outlets, and negative sentiment terms are prominent.

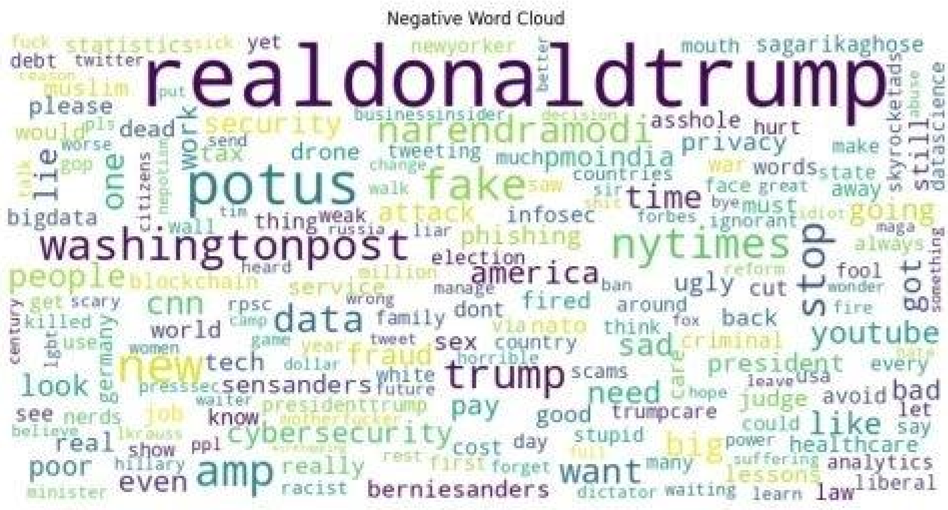

Figure: Word Cloud — Negative Tweets (Google Colab)

- Dominant words: 'realdonaldtrump', 'potus', 'washingtonpost', 'nytimes', 'fake', 'data'
- Strong presence of political and media-related terms indicating political discourse.
- Words like 'fake', 'lie', 'security', 'phishing' suggest disinformation-related tweets.
- Color scheme uses cool greens and purples consistent with negative sentiment visualization.

Positive Tweets — Word Cloud

The Positive Word Cloud highlights the most frequently occurring words in tweets classified with positive sentiment. Terms related to technology, data science, and positive endorsements are prominently featured.

Figure: Word Cloud — Positive Tweets (Google Colab)

- Dominant words: 'realdonaldtrump', 'narendramodi', 'nytimes', 'amp', 'security', 'blockchain'
- Technology terms like 'blockchain', 'bigdata', 'analytics', 'machinelearning' are highly visible.
- Positive action words: 'great', 'best', 'thank', 'congratulation', 'hope', 'innovation'

- Color scheme uses warm greens and teals indicating optimistic sentiment.

Key NLP Insights

- Both clouds share 'realdonaldtrump' and 'narendramodi' — public figures dominate Twitter discourse regardless of sentiment.
- Negative tweets lean toward political criticism and security concerns.
- Positive tweets lean toward technology, data science, and professional achievements.
- The word cloud visualization provides an intuitive, visual summary of text data that supplements numerical analysis.

End of Assignment | Submitted by: Ammar | Course: Machine Learning / AI